MAHATMA GANDHI UNIVERSITY

MGU-UGP (HONOURS)

THIRD SEMESTER

Discipline Specific Core Course (DSC) – MG3DSCMBG 201 – MICROBIAL GENETICS

(2024 ADMISSION ONWARDS)

Duration: 1.5 Hours Maximum marks: 50

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I)and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overall

Outcome attainability.

PART A

Short Answer type questions, Answer any 10

Each question carries 2 marks

1. How does an episome differ from other plasmids in bacteria?[U] / [CO1]

2.Differentiate between the leading strand and the lagging strand in

DNA replication[U] / [CO1]

3.What are F+ cells [K] / [CO2]

4. What do you mean by T-even phages? [K] / [CO2]

5.Give an account ofthe Ecoli chromosome[U] / [CO1]

6.What is the Pribnow box, and where is it located in prokaryotic DNA [K] / [CO2]

7.What is an aporepressor[K] / [CO2]

8.Differentiate between auxotrophs and prototrophs[U] / [CO3]

9. What are competent cells in bacteria, and why are they important in genetic transformation? [U] / [CO2]

10.What is photo-reactivation repair? [K] / [CO3]

11.What is induction in the control of gene expression in prokaryotes [K]/ [CO2]

12.What is a Col plasmid? State its function. [K] / [CO1]

PART B

Short Essay questions, Answer any 4

Each question carries 5 marks.

13. How did Harshey and Chase prove that DNA is the genetic material?[U]/[CO1]
14. Explain the cut-and-paste mechanism of transposition[U]/ [CO1]
15. Illustrate and explain Translation in prokaryotes.[A]/CO2]
16. What are the steps involved in transcription [U] / [CO 2]
17. Explain the Lac operon in detail.[U]/ [CO2]
18. Explain the principle and significance of the Ames test in detecting mutagenic compounds.[U] / [CO3]

Part C

Long Essay Question. Answer any 1 question

Each question carries 10 marks

19. Analyse how different horizontal gene transfer mechanisms contribute to the transmission of drug resistance in bacteria [An]/ [CO2]

20. Write an essay on DNA repair mechanisms.[An]/ [CO3]